SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: DSA0204**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage:20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection
A city surveillance system must operate under the following conditions:

- **If latency > 100 ms → system fails**
- **If accuracy < 85% → unacceptable**
- **Hardware supports only moderate computation**

Models available:

Viola-Jones: 78%, low latency
YOLO: 92%, moderate latency
Deep Learning: 95%, high latency

Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

1. Smart Surveillance System Selection
Given constraints:

- Latency must be ≤ 100 ms.
- Accuracy must be ≥ 85%.
- Hardware supports only moderate computation.

Model	Accuracy	Latency	Computation	Constraint Result
Viola-Jones	78%	Low	Low	Accuracy fails
YOLO	92%	Moderate	Moderate	Satisfies all
Deep Learning	95%	High (>100 ms)	High	Latency & hardware fail

Evaluation:

- Viola-Jones has low latency and low computation, but its accuracy is only 78%, which is below the required 85%.
- Deep Learning provides the highest accuracy, but its high latency violates the 100 ms requirement and it requires more computation than the available hardware supports.
- YOLO provides 92% accuracy, moderate latency, and moderate computational requirements.

Decision:

YOLO should be selected.

Justification:

YOLO is the only model that satisfies all three constraints simultaneously: accuracy ≥85%, latency ≤100 ms, and moderate computational requirements.

2. Autonomous Vehicle Detection Logic

A vehicle detection system must:

- **Prioritize safety (accuracy > 90%)**
- **Maintain response time < 50 ms**

Options:

Method A: 88% accuracy, 30 ms

Method B: 93% accuracy, 70 ms

Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

2. Autonomous Vehicle Detection Logic

Given constraints:

- Accuracy must be >90% for safety.
- Response time must be <50 ms.

Method	Accuracy	Response Time	Result
Method A	88%	30 ms	Accuracy fails

Method B	93%	70 ms	Response time fails
Method C	91%	45 ms	Both satisfied

Evaluation:

- Method A has excellent response time, but 88% accuracy is below the safety requirement.
- Method B has high accuracy of 93%, but its 70 ms response time exceeds the 50 ms limit.
- Method C provides 91% accuracy and 45 ms response time, satisfying both requirements.

Decision:
Method C should be selected.
Justification:
In an autonomous vehicle, both safety and fast response are essential. Method C is the only option that meets both constraints, making it the safest and most suitable choice**.**

3. Motion Analysis Decision Problem
A traffic system requires:

- **Accurate motion direction detection**
- **Robustness in dense traffic**

Results:

Optical Flow: accurate but noisy in dense scenes
Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

3. Motion Analysis Decision Problem

Requirements:

- Accurate motion-direction detection.
- Robust operation in dense traffic.

Approach	Accuracy/Detail	Dense Traffic Robustness	Evaluation
Optical Flow	Accurate and detailed	Noisy	Less suitable
Motion Estimation	Less detailed	Stable	More suitable

Evaluation:

- Optical Flow can provide accurate and detailed motion information. However, in dense traffic, multiple moving objects can produce overlapping motion patterns, making the result noisy.
- Motion Estimation provides less detailed information but is more stable when many vehicles are present.

- Since the system specifically requires robustness in dense traffic, stability is more important than maximum detail.

Decision: Motion Estimation should be selected.

Justification: Under high-density traffic conditions, a stable and reliable method is preferred. Motion Estimation can handle crowded scenes better and reduce the effect of noise.

4. Background Modeling Evaluation
A GMM-based system produces:

- **Static scenes: 90% accuracy**
- **Dynamic scenes: 65% accuracy**

Requirement:

System must maintain ≥80% accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

Requirement:

- Accuracy must be ≥80% in all scenarios.

Scenario	GMM Accuracy	Required Accuracy	Result
Static scenes	90%	≥80%	Pass
Dynamic scenes	65%	≥80%	Fail

Evaluation:

- GMM performs well in static scenes with 90% accuracy.
- However, its performance decreases to 65% in dynamic scenes because moving backgrounds, shadows, lighting changes, and other variations can affect background modeling.
- Since the system must maintain at least 80% accuracy across all scenarios, GMM does not satisfy the complete requirement.

Decision:

GMM alone should not be selected.

Suggested approach: Improve GMM using adaptive background modeling, shadow removal, or a more robust background-subtraction technique.

Justification:

Although GMM performs well in static environments, its 65% accuracy in dynamic scenes violates the minimum 80% requirement. Therefore, additional techniques are necessary.

5. Depth Estimation Strategy

Constraints:

- **Only one camera available**
- **Moderate depth accuracy acceptable**

Options:

Stereo Vision: high accuracy, needs two cameras

Structure from Motion: moderate accuracy, single camera

Compare both approaches and evaluate which is feasible. Justify using constraint-based reasoning.

5. Depth Estimation Strategy

Given constraints:

- Only one camera is available.
- Moderate depth accuracy is acceptable.

Method	Cameras Required	Accuracy	Constraint Result
Stereo Vision	2 cameras	High	Not feasible
Structure from Motion	1 camera	Moderate	Feasible

Evaluation:

- Stereo Vision can provide high-quality depth information by comparing images from two cameras. However, the system has only one camera, so this method cannot satisfy the hardware constraint.
- Structure from Motion (SfM) estimates depth from multiple views captured using a single moving camera.
- Since only moderate depth accuracy is required, SfM provides a suitable balance between feasibility and accuracy.

Decision:

Structure from Motion should be selected.

Justification: SfM requires only one camera and provides moderate depth estimation, which matches the given requirements. Therefore, it is the most feasible approach.

Final Decisions

Problem	Selected Technique	Main Reason
1. Smart Surveillance	YOLO	Meets accuracy, latency, and computation constraints
2. Autonomous Vehicle	Method C	Accuracy >90% and response time <50 ms
3. Motion Analysis	Motion Estimation	More robust in dense traffic
4. Background Modeling	GMM + improvement	GMM alone fails in dynamic scenes
5. Depth Estimation	Structure from Motion	Works with a single camera and moderate accuracy

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Code of Conduct:

I M Jeevitha (192424257) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Jeevitha*

RUBRICS FOR CALCULATION:

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation